Slope, Revenue & Area Drill

Slope · Revenue · Area as a Function · Logarithms

PROBLEM 1 — INTERPRETING SLOPE ON A MOUNTAIN ROAD

PROBLEM 1

A mountain road drops **42 feet** in elevation for every **700 feet** of horizontal distance.

RECALL
Percent grade = (vertical drop ÷ horizontal distance) × 100%

(a) What is the percent grade of this road? Show your calculation.

work

Grade = ______ %

(b) Write a linear equation for vertical drop V (feet) as a function of horizontal distance H (feet).

work

$V =$ ______

(c) A driver starts at the top. After driving some distance, the altimeter shows the car is **84 feet lower** than the start. How far has the car traveled horizontally?

work

$H =$ ______ feet

(d) The base of the hill is **2,500 feet** horizontally from the top. What is the total height of the hill?

work

Height = ______ feet

PROBLEM 2 — REVENUE AS A FUNCTION

PROBLEM 2

A streaming service charges **$150 − 7x\)** dollars per subscription, where x is the number of subscriptions sold (in thousands). Revenue is price times quantity, so $R(x) = x(150 - 7x)$, measured in thousands of dollars.

(a) Expand $R(x)$ and write it in standard form $ax^2 + bx + c$. Identify a, b, and c.

work

$R(x) =$ __________

$a =$ $b =$ $c =$ __________

(b) What values of x make $R(x) = 0$? What do those values mean in context?

work

$x =$ __________

Meaning: __________

(c) Use the vertex formula $x = -b/(2a)$ to find the subscription count that **maximizes revenue**. Leave your answer as a fraction.

work

$x =$ __________ thousand subscriptions

(d) Calculate R at that maximum. Round to the nearest whole number (in thousands of dollars). What is the maximum revenue?

work

Max $R \approx$ __________ thousand dollars

(e) The company needs $R(x) = 700$ (thousand dollars). Set up the equation and solve using the quadratic formula. There are two solutions — what does each one mean for the business?

work

$x_1 \approx$ ______ thousand

$x_2 \approx$ ______ thousand

Meaning: ______

PROBLEM 3 — TRIANGLE UNDER A LINE

PROBLEM 3

The line $f(x) = 48 - 6x$ is drawn on a coordinate plane. A right triangle is formed with its base along the x-axis from 0 to x, and its height going straight up from $(x, 0)$ to the point $(x,\ f(x))$ on the line.

(a) Write the area function $A(x)$ using the formula $A = \frac{1}{2} \cdot \text{base} \cdot \text{height}$. Simplify completely.

work

$A(x) =$ ______

(b) What is the domain of $A(x)$? Explain why x cannot go higher than its upper bound.

work

Domain: ______

(c) Complete the table.

x	1	2	3	4	5	6	7
$A(x)$							

What pattern do you notice in the table?

observation

(d) Use the vertex formula to find the value of x that maximizes the area. What is the maximum area?

work

$x =$ ______

Max area = ______ square units

(e) Find all values of x where $A(x) = 45$. Factor to solve — no quadratic formula needed. Describe the two triangles.

work

$x =$ ______ and ______

(f) Find all values of x where $A(x) = 36$. For each solution, write the base and height of the triangle and explain why the two triangles have the same area even though they look different.

work

$x =$ ______ and ______

Explanation: ______

PROBLEMS 4–7 — EVALUATING LOGARITHMS

PROBLEM 4

Evaluate: $\log_{27}(9)$

(a) Write 27 and 9 as powers of a common base.

work

$27 =$ ______ $9 =$ ______

PROBLEM 5

Evaluate: $\log_{32}(8)$

(a) Write 32 and 8 as powers of a common base.

work

$32 =$ ______ $8 =$ ______

(b) Rewrite $27^y = 9$ using your common base. Set the exponents equal and solve for y.

work

$\log_{27}(9) =$ ____

(b) Rewrite $32^y = 8$ using your common base. Set the exponents equal and solve for y.

work

$\log_{32}(8) =$ ____

PROBLEM 6

Evaluate: $\log_{16}(32)$

(a) Write 16 and 32 as powers of a common base.

work

$16 =$ ____ $32 =$ ____

(b) Rewrite $16^y = 32$ using your common base. Set the exponents equal and solve for y.

work

$\log_{16}(32) =$ ____

PROBLEM 7

Evaluate: $\log_{125}(25)$

(a) Write 125 and 25 as powers of a common base.

work

$125 =$ ____ $25 =$ ____

(b) Rewrite $125^y = 25$ using your common base. Set the exponents equal and solve for y.

work

$\log_{125}(25) =$ ____